

NIEDERSTETTEN, Germany

WMO: 107430

Lat: 49.3894N

Lon: 9.9667E

Elev: 1506

StdP: 13.91

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.5	(c) 17.1	(d) 8.4	(e) 9.0	(f) 14.1	(g) 13.0	(h) 11.3	(i) 19.6	(j) 34.1	(k) 43.4	(l) 29.8	(m) 42.7	(n) 7.5	(o) 50	(p) 0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	86.6	66.6	82.7	65.1	79.2	64.0	68.2	82.3	66.7	79.7	65.2	77.0	6.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	92.9	72.7	62.2	88.6	71.5	60.7	83.9	69.8	33.3	82.3	32.1	79.5	30.9	76.8	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.1	(b) 20.9	(c) 17.5	(e) 8.2	(f) 92.2	(g) 5.2	(h) 3.1	(i) 4.5	(j) 94.5	(k) 1.4	(l) 96.3	(m) -1.5	(n) 98.1	(o) -5.3	(p) 100.3		
			DB	7.7	71.5	5.2	2.0	3.9	72.9	0.8	74.1	-2.1	75.1	-5.9	76.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	31.9	34.1	40.8	48.3	55.8	61.9	65.3	65.0	57.2	49.4	40.0	34.3
	DBStd	13.89	8.36	8.29	7.44	7.64	6.97	6.96	6.54	6.56	6.27	7.00	7.01	7.66
	HDD50	2355	562	447	295	122	24	1	0	0	9	98	308	489
	HDD65	6178	1027	866	750	502	297	141	79	82	246	484	751	953
	CDD50	1890	0	1	10	71	203	358	474	465	223	79	6	0
	CDD65	242	0	0	0	0	11	49	88	82	11	1	0	0
	CDH74	2289	0	0	0	9	111	470	833	756	106	4	0	0
	CDH80	672	0	0	0	1	15	122	269	249	16	0	0	0
	WSAvg	7.9	9.6	9.2	9.2	7.5	7.0	6.6	6.8	6.6	6.9	7.6	8.3	9.1
	PrecAvg	30.3	2.4	2.3	2.4	1.9	2.6	2.7	3.0	2.5	2.4	2.6	2.6	2.8
(15) Precipitation	PrecMax	40.0	5.1	6.3	6.1	5.2	5.7	6.4	5.4	5.5	5.2	8.1	4.8	8.0
	PrecMin	20.4	0.4	0.6	0.3	0.1	0.6	1.0	0.6	0.5	0.8	0.7	0.1	0.4
	PrecStd	4.4	1.2	1.4	1.5	1.1	1.3	1.2	1.3	1.3	1.1	1.5	1.2	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	57.4	64.7	75.0	82.0	88.5	91.6	92.8	82.4	73.1	60.3	52.2
		MCWB	46.9	47.7	51.1	57.1	64.5	68.4	67.6	66.7	63.7	60.0	53.8	47.0
	2%	DB	49.3	51.5	60.4	70.0	77.1	83.5	86.6	86.3	77.1	66.6	55.2	49.7
		MCWB	45.6	44.8	49.2	54.5	62.4	66.3	66.4	66.2	62.1	58.0	50.5	46.5
	5%	DB	46.1	48.0	55.6	66.0	72.7	79.2	82.6	82.0	72.6	62.7	52.1	47.0
		MCWB	43.1	43.5	46.7	53.0	59.2	64.8	64.9	64.7	60.4	56.2	48.4	44.6
	10%	DB	43.0	44.9	51.7	61.5	68.8	75.5	78.7	77.6	68.2	59.2	49.7	44.5
		MCWB	40.8	41.3	45.6	50.9	57.2	63.1	63.7	63.4	58.7	54.4	46.8	42.6
	0.4%	WB	48.9	48.5	52.9	58.4	66.4	71.1	69.8	69.5	65.5	61.7	54.5	49.3
		MCDB	51.1	54.7	62.5	71.1	78.9	83.5	85.4	84.6	77.1	69.5	58.3	50.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.8	46.0	50.1	55.8	63.4	68.0	67.9	67.7	63.2	58.9	51.3	47.0
		MCDB	48.8	50.2	56.5	67.6	74.7	80.0	82.8	81.9	73.8	65.0	54.5	48.9
	5%	WB	43.0	43.9	48.2	53.8	60.9	65.7	66.3	66.1	61.5	56.9	49.1	44.5
		MCDB	45.3	47.3	54.0	63.9	70.5	77.1	79.3	79.4	70.6	61.7	51.6	46.8
	10%	WB	40.6	41.5	46.0	51.6	58.3	63.7	64.8	64.4	59.6	54.9	46.9	42.4
		MCDB	42.6	44.4	51.0	60.6	67.1	74.3	76.3	75.9	67.0	58.9	49.3	44.1
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	10.9	13.5	16.9	17.5	18.1	18.1	18.0	15.9	13.1	9.1	8.0
		MCDBR	11.6	15.6	20.4	23.8	23.6	24.5	25.3	25.0	22.6	18.2	13.7	10.6
		MCWBR	9.2	10.8	12.1	12.5	12.5	12.0	10.5	10.4	11.0	10.9	9.5	8.4
	5% WB	MCDBR	10.7	13.7	17.2	21.6	21.6	22.7	21.6	22.2	19.4	16.3	12.4	9.8
		MCWBR	9.0	10.1	11.0	11.9	12.1	11.7	9.6	10.2	10.2	10.3	9.2	8.2
(40) Clear-Sky Solar Irradiance	taub		0.299	0.323	0.356	0.387	0.395	0.398	0.403	0.398	0.369	0.347	0.322	0.298
	taud		2.438	2.388	2.329	2.270	2.271	2.290	2.269	2.309	2.370	2.429	2.454	2.436
	Ebn at Noon		241	261	270	272	274	273	270	266	262	248	229	224
	Edh at Noon		20	26	33	39	41	40	40	37	31	25	20	17
	RadAvg		272	525	886	1357	1605	1789	1691	1467	1072	635	311	214
(45) All-Sky Solar Radiation	RadStd		43	86	120	179	178	145	164	138	134	71	36	27

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.88	N/A	N/A	N/A	N/A	N/A	N/A	-270	+131	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page